example there is a good two-dimensional stand-in: an infinite version of a Pringle's potato chip. This shape, often referred to as a *saddle*, is a kind of inverse of the sphere: Whereas a sphere is symmetrically bloated outward, the saddle is symmetrically shrunken inward, as illustrated in Figure 8.6. Using a bit of mathematical terminology, we say that the sphere is *positively curved* (bloats outward), the saddle is *negatively curved* (shrinks inward), and flat space—whether infinite or finite—has *no curvature* (no bloating or shrinking).[21]

Researchers have proven that this list—uniformly positive, negative, or zero—exhausts the possible curvatures for space that are consistent with the requirement of symmetry between all locations and in all directions. And that is truly stunning. We are talking about the shape of the *entire universe*, something for which there are endless possibilities. Yet, by invoking the immense power of symmetry, researchers have been able to narrow the possibilities sharply. And so, if you allow symmetry to guide your answer, and your late-night interrogator grants you a mere handful of guesses, you'll be able to meet his challenge.[13]

All the same, you might wonder why we've come upon a variety of possible shapes for the fabric of space. We inhabit a single universe, so why can't we specify a unique shape? Well, the shapes we've listed are the only ones consistent with our belief that every observer, regardless of where in the universe they're located, should see on the largest of scales an identical cosmos. But such considerations of symmetry, while highly selective, are not able to go all the way and pick out a unique answer. For that we need Einstein's equations from general relativity.

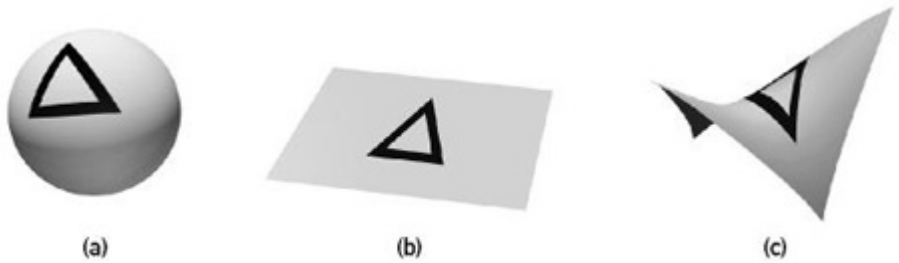


Figure 8.6 Using the two-dimensional analogy for space, there are three types of curvature that are completely symmetric—that is, curvatures in which the view from any location is the same as that from any other. They are **(**a**)** *positive* curvature, which uniformly bloats outward, as on a